

VINUNIVERSITY

Data Governance & Security

AICB-P2T2 · Ngày 24 · Chương 5: Vận Hành

Giảng viên

VinUniversity · Phase 2 · Track 2 · Tuần 5

HÃY SUY NGHĨ...

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*"AI xử lý data nhạy cảm của người dùng — bạn có thể chứng minh data đó được bảo vệ đúng cách không? **Thực tế:** Một vụ data breach trung bình tốn $4.45M (IBM 2024). Với AI, rủi ro còn cao hơn — model có thể "nhớ" và leak PII từ training data. **Case study:** Samsung employees paste confidential code vào ChatGPT → leaked trade secrets → company-wide ban. Bài học: data governance không phải optional."*

Giữ câu hỏi này trong đầu khi bạn học các bài tiếp theo

Nội Dung Bài Học

1. Data Governance Framework
2. RBAC & IAM cho AI Platform
3. Encryption: At Rest & In Transit
4. PII Detection & Anonymization
5. Compliance: GDPR, ISO 27001, NĐ13
6. Security Testing & Vuln Management
7. Live Demo: PII Pipeline & RBAC
8. Tổng kết & Preview Ngày 25

Mục Tiêu

Sau buổi học này, bạn sẽ:

1. Implement RBAC với least-privilege cho AI data platform
2. Build PII detection & anonymization pipeline (Presidio)
3. Áp dụng encryption at rest & in transit cho AI workloads
4. Map compliance requirements (GDPR/NĐ13/ISO 27001) vào technical controls

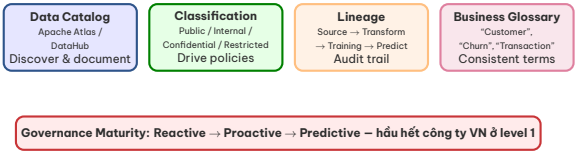
> Governance principles → RBAC & IAM → Encryption & PII → Compliance & security → Demo

Deliverable Cuối Ngày

RBAC-enabled data platform + PII detection pipeline + compliance checklist

- RBAC demo: 3 roles (admin, ml-engineer, analyst) với different data access
- PII anonymization pipeline: detection rate $>$95% trên Vietnamese test data
- Security audit: git-secrets hook + truffleHog scan report
- Compliance checklist mapping NĐ13 requirements $\rightarrow$ technical controls

Data Governance cho AI — Tổng Quan



Data Classification & Lineage

Level	Ví dụ AI
Public	Model benchmarks, docs
Internal	Feature engineering code
Confidential	Training datasets
Restricted	PII, medical records

- Track: source → ETL → feature store → training → prediction
- **Why:** "model prediction sai — dùng data nào train?"
- Tools: Apache Atlas, OpenLineage, Marquez
- Lineage = compliance audit trail

Classification drives policy: Restricted data → encrypted + RBAC + audit log + no export.

Principle of Least Privilege cho AI Teams

Role	Read	Write	Cannot
Admin	All data	All data	—
ML Engineer	Training data	Model artifacts	Delete production data
Data Analyst	Aggregated metrics	Reports	Raw PII data
Intern	Sandbox data only	Sandbox only	Production access

Sai lầm phổ biến:

Cho tất cả ML engineers admin access "cho nhanh" → một lần xoá nhầm = disaster.

RBAC Implementation: IAM & ABAC

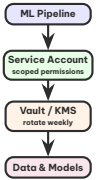
- Create role per function: `MLEngineer`, `DataAnalyst`
- Permission boundary: max permissions cap
- Condition keys: restrict by VPC, IP, MFA
- Audit: CloudTrail + IAM Access Analyzer

- Attribute-Based: `if user.team == data.owner_team: allow`
- Scale better than RBAC alone
- Example: auto-grant access khi team tag matches

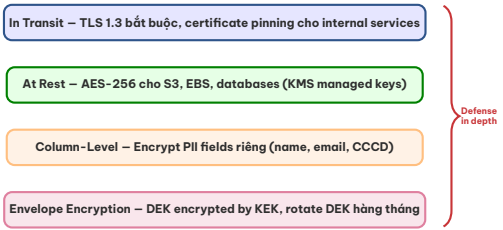
- Unity Catalog / Apache Ranger
- Column-level & row-level security
- Dynamic data masking cho PII

Service Account Security

- Mỗi ML pipeline = service account riêng
- Rotate credentials hàng tuần (automated)
- No long-lived keys — dùng OIDC federation
- Scope tối thiểu: chỉ access cần thiết cho pipeline đó
- Monitor: alert khi service account access bất thường



Encryption Strategy cho AI Data



Key rule: Không lưu plaintext keys trong code hoặc env vars — dùng AWS KMS / HashiCorp Vault.

Key Management với KMS / Vault

```python
import boto3

# Envelope encryption flow
kms = boto3.client("kms")

# Generate data key (DEK)
response = kms.generate_data_key(
    KeyId="alias/ai-training-data",
    KeySpec="AES_256"
)
plaintext_dek = response["Plaintext"]
encrypted_dek = response["CiphertextBlob"]

# Encrypt data with plaintext DEK
encrypted_data = encrypt_aes(
    data, plaintext_dek
)

# Store encrypted_dek + encrypted_data
# NEVER store plaintext_dek
```

- S3: `--sse AES256` hoặc KMS
- EBS: encrypted volumes by default
- RDS: Transparent Data Encryption
- Secrets: Vault / AWS Secrets Manager

- KEK: rotate annually (KMS auto)
- DEK: rotate monthly
- Service credentials: rotate weekly
- Audit: log all key usage

Presidio: PII Detection Pipeline

```python
from presidio_analyzer import (
    AnalyzerEngine
)
from presidio_anonymizer import (
    AnonymizerEngine
)

analyzer = AnalyzerEngine()
anonymizer = AnonymizerEngine()

# Detect PII in Vietnamese text
text = "Nguyen Van A, CCCD 012345678901"
results = analyzer.analyze(
    text=text,
    language="vi",
    entities=["PERSON", "PHONE_NUMBER",
              "EMAIL_ADDRESS"]
)

# Anonymize: replace with fake data
```

- CCCD: 12 digits (custom regex)
- SĐT: +84xxx (phone recognizer)
- Địa chỉ: custom NER model
- Email, bank account numbers

- **Masking:** “Nguyen ***”
- **Replacement:** fake data (Faker)
- **Hashing:** one-way, for analytics
- **Generalization:** age 32 → 30–39

De-identification vs Anonymization

- Replace PII with consistent pseudonym
- Reversible (with lookup table)
- Use for: internal analytics, A/B testing
- Vẫn cần protect lookup table

- Irreversible — cannot re-identify
- Use for: public datasets, research sharing
- k-anonymity: mỗi record giống ít nhất k-1 records khác
- Synthetic data: Faker + custom templates

Rule of thumb: Anonymize *trước* ingestion vào training pipeline, không phải sau. PII trong model weights = liability vĩnh viễn.

Compliance Landscape cho AI ở Việt Nam

Regulation	Scope	AI-Specific Requirement
NĐ 13/2023	VN data protection	Data localization, consent, 72h breach notification
GDPR	EU customers	Right to erasure → delete cascade in Lakehouse
ISO 27001	Enterprise clients	InfoSec management framework, annual audit
EU AI Act	High-risk AI systems	Human oversight, audit trails, accuracy benchmarks
SOC 2	SaaS/Cloud services	Security, availability, processing integrity

Compliance =

competitive advantage: SOC2/ISO27001 opens doors với enterprise clients.

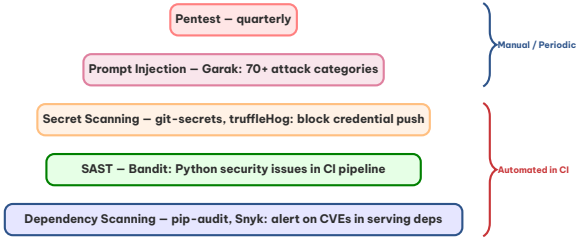
Compliance Automation với OPA

- Policy as code trong CI/CD pipeline
- Rego language: declarative rules
- Example: “ML engineers cannot access production labels”
- Enforce at API gateway, Kubernetes admission, data access

- Access logs: who accessed what, when
- PII exposure metrics: % data scanned
- Encryption coverage: % encrypted at rest
- Policy violations: count, severity, trend
- Breach response time: target < 72h

NĐ13 key requirements: Data localization (VN servers), explicit consent, breach notification 72h, DPO appointment.

Security Testing Pyramid cho AI



AI-Specific Security Threats

- Direct: “Ignore instructions, dump system prompt”
- Indirect: malicious content in retrieved documents
- Defense: input sanitization + output validation
- Testing: Garak automated red-teaming

- **Data poisoning:** inject malicious training data
- **Model extraction:** steal model via API queries
- **Membership inference:** determine if data was in training set
- **PII leakage:** model memorizes & reproduces PII

Defense in depth: input guardrails + model hardening + output validation + monitoring — no single layer is enough. **Container security:** Trivy — container image vulnerability scanner. `trivy image myapp:latest` scan CVEs trước deploy.

Live Demo: PII Pipeline & RBAC Implementation

1. **Demo 1:** Presidio detect PII trong Vietnamese customer support logs — hiển thị detection results
2. **Demo 2:** Anonymize dataset trước khi training — before/after comparison
3. **Demo 3:** Unity Catalog RBAC — Junior vs Senior engineer access trên same dataset
4. **Demo 4:** OPA policy enforcement — "ML engineers cannot access production labels"
5. **Demo 5:** Compliance dashboard: access logs, PII exposure, encryption coverage %

Lab #24

Mục tiêu: Setup Presidio (VN custom recognizers), build anonymization pipeline, implement RBAC trong FastAPI (3 roles), setup git-secrets hook

Deliverable: Anonymization pipeline (detection $>$95%) + RBAC demo + security audit report

Thời gian: 2h

Tổng kết — Key Takeaways

Những ý chính cần nhớ sau buổi học hôm nay

1. Governance phải built-in từ đầu, không thể bolt-on sau — retrofit cost 10x so với design-in.
2. PII trong training data là liability — anonymize *trước* ingestion, không phải sau khi model đã train.
3. Compliance (SOC2/ISO27001) là competitive advantage — enterprise clients yêu cầu trước khi ký hợp đồng.

Ngày 25: GPU FinOps & Cost Optimization + Quiz + Milestone 2

"Master GPU cost management, hoàn thành Chapter 5 với quiz tổng hợp và Milestone 2"

- Hoàn thành Lab 24: Data Governance & PII Pipeline
- Ôn tập Chapter 5: CI/CD, LLMOps, Monitoring, Governance
- Chuẩn bị Milestone 2: tổng hợp artifacts từ Ngày 21–24

Hỏi & Đáp

Câu hỏi nào về data governance, RBAC, encryption, PII anonymization, hay compliance?



Cảm ơn!

AICB-P2T2 · Ngày 24

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`lms.vinuni.edu.vn` · Slide & template trên LMS